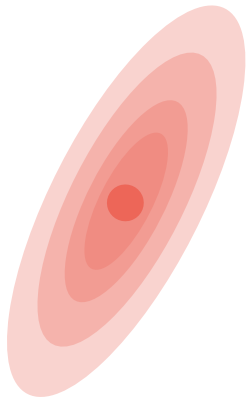


consensus meeting point

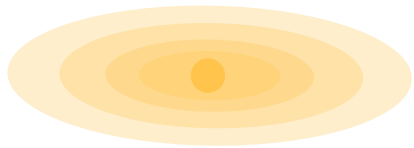
ρ_1



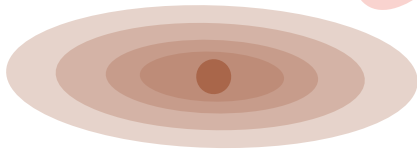
ρ_2



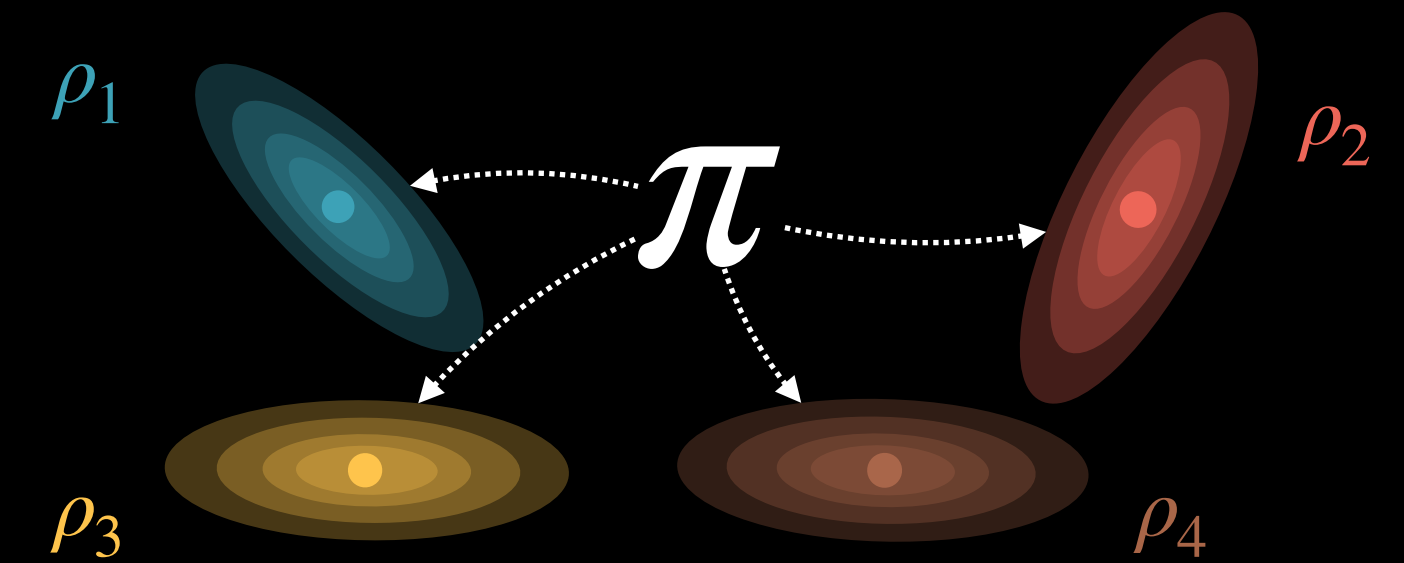
ρ_3



ρ_4



Multi-Marginal OT: Intuition



- Generalizes OT from two to many distributions: finds a joint coupling linking all marginals simultaneously.
- Coupling acts as “**consensus**” or “**meeting point**”
- Value of $\pi(x_1, \dots, x_S)$ represents consensus mass for tuple $\pi(x_1, \dots, x_S)$ among all sources.
- The solution balances transport so that all marginals “agree” on where their mass should meet.
- For Wasserstein barycenters (special case) the ‘meeting point’ defines the barycenter measure.

Multi-Marginal OT: Computation